

# Common Ownership's Effect on Domestic Airline Productivity

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# Background

- In the past 20 years, shares owned by institutional investors has risen from 33% to 61%
- We propose the following:
  - ① An increase in common ownership incentivizes the sharing of technological innovations among competitors
  - ② Common ownership also encourages collusive behavior, resulting in higher consumer prices & decreased investment in cost reduction

## Related Literature

- Alam & Sickles (2000) - Looks at productivity, efficiency, and technological change in airlines
- Bas, Demmou, Franco, & Garcia-Bernardo (2023) - Examines how ownership affects firm level productivity across 18 OECD countries
- Olley & Pakes (1996) and Levinsohn & Petrin (2003) - Propose methods for estimating productivity using intermediates to recover productivity
- Azar, Schmalz, Tecu (AST, 2018) and Dennis, Gerardi, & Schenone (2022) - Examines how ownership affects pricing in the airline industry

# Innovation

- The paper written by Dennis, Gerardi, and Schenone (DGS) expanded on AST, examining more closely the relationship between common ownership and anti-competitive pricing
- DGS looked at both carrier and market level data, finding that in both instances the observed correlation was a result of market concentration, not common ownership

## Relevant Connection

We consider the DGS paper in two facets:

- ① First, DGS highlights the necessary (but not sufficient) conditions that, if there is anti-competitive behavior, then there must be correlation between the ownership metric and competitive behavior
- ② Second, DGS only establishes that common ownership may not have a causal relationship with pricing behavior, however neither paper considers how common ownership may disincentivize technological innovation by reducing competitive pressure

# Implications

- Since DGS only shows that common ownership does not cause anti-competitive pricing behavior, we consider whether common ownership has anti-competitive effects on technological innovation
- Thus, first we construct a proxy for technological innovation
- Note that, unlike DGS, we only consider carrier-level data, out of necessity

# Total Factor Productivity

- To measure technological innovation, we use Total Factor Productivity (TFP) as a proxy
- In order to estimate TFP we use a procedure described in a paper by Levinsohn & Petrin (2003), in which they derive an estimation procedure similar to that of Olley & Pakes (1996), but employ material costs rather than investment

## LP Model: Production Function

The model assumes a Cobb-Douglas production function with unobserved productivity ( $\omega_{it}$ ). Applying a log-transform yields:

$$\ln(Y_{it}) = \beta_l \ln(L_{it}) + \beta_k \ln(K_{it}) + \omega_{it} + \epsilon_{it}$$

- $\ln(Y_{it})$ : Real Revenue
- $\ln(L_{it})$ : Real Labor (free input)
- $\ln(K_{it})$ : Real Capital (semi-fixed)

### Endogeneity Problem

- Productivity ( $\omega_{it}$ ) is correlated with choices of labor and capital, so direct estimation  $\implies$  error term correlation
- This correlation leads to *simultaneity bias*

## LP Model: Identification

To address the simultaneity bias of  $\omega_{it}$ , it is modeled as a nonparametric function of observable inputs:

$$\omega_{it} = h(\ln(K_{it}), \ln(M_{it}))$$

Where  $\ln(M_{it})$  is Real Fuel Cost, an observable intermediate assumed to be *monotonically increasing* in  $\omega_{it}$  (more productive firms use more fuel). Thus we can invert the function and express  $\omega_{it}$  as a function of observable inputs

### Estimation Procedure

- 1 Estimate coefficients of the free input (Labor) using the proxy function
- 2 Estimate the fixed input coefficients (Capital) and correct for survival bias

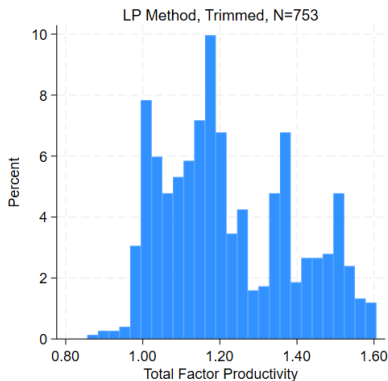
# TFP Estimation

As described in LP, we used the following data:

- ① Output - Operating Revenue, Form 41 Schedule P-1.2
  - Deflator - PPI Scheduled Passenger Air Transport
- ② Capital - Aircraft Equipment, Form 41 Schedule B-1
  - Deflator - PPI Transportation Equipment: Aircraft
- ③ Labor - Salary & Fringe Benefits, Form 41 Schedule P-6
  - Deflator - Average Hourly Earnings for Transportation Workers
- ④ Materials - Domestic Fuel Cost, Form 41 Schedule P-12(a)
  - Deflator - PPI Jet Fuel

# TFP Estimation Results

## Distribution of TFP Estimates



## Descriptive Statistics

|                      |              |
|----------------------|--------------|
| Observations ( $N$ ) | 767          |
| <b>Mean</b>          | <b>1.224</b> |
| <b>Median</b>        | <b>1.187</b> |
| Std. Dev.            | 0.182        |
| Skewness             | 0.157        |

| <b>Percentile</b> | <b>TFP</b> |
|-------------------|------------|
| 5%                | 0.998      |
| 95%               | 1.580      |

# Common Ownership

The  $\text{MHHi}\Delta$  for firm  $i$  measures the degree to which firm  $i$  and its rivals  $j$  are owned by the same investor:

$$\text{MHHi}\Delta_{it} = \sum_{j \neq i}^N \left( s_{jt} \cdot \sum_k \alpha_{ikt} \alpha_{jkt} \right)$$

- $s_{jt}$ : Market share of firm  $j$ .
- $k$ : Institutional investor
- $\alpha_{ikt}$ : Ownership stake of investor  $k$  in firm  $i$ .
- $(\sum_k \alpha_{ikt} \alpha_{jkt})$ : The *commonality* between firm  $i$  and rivals  $j$ .

## Key Point

- Investors will use control in firm  $i$  to make it compete less with its rival firm  $j$
- Thus we postulate that  $\uparrow \text{MHHi}\Delta \implies \downarrow \text{TFP}$

# Ownership Calculation

We use the SEC Form 13F dataset to calculate MHHI $\Delta$  as:

- 1 Market Share ( $s_{jt}$ ) - Percentage of domestic air transport by carrier  $j$  in period  $t$
- 2 Ownership Stake ( $\alpha_{ikt}$ ) - Percentage of share of firm  $i$  owned by investor  $k$

Note that DGS considers market (route) level market share, whereas our analysis is independent of route and only considers different carriers

## Model Specification

We model our estimated firm-level TFP ( $\hat{\omega}_{it}$ ) using the following specification:

$$\hat{\omega}_{it} = \beta_0 + \beta_1 MHHI\Delta_{it} + \gamma X_{it} + \mu_i + \delta_t + \varepsilon_{it}$$

- $\hat{\omega}_{it}$ : Estimated TFP
- $X_{it}$ : Controls (e.g. size, HHI)
- $\beta_1$ : Coefficient of interest
- $\mu_i$ : Firm Fixed Effects
- $\delta_t$ : Time Fixed Effects
- $\varepsilon_{it}$ : Error Term

### Hypothesis

If common ownership leads to reduced competitive pressure, the expected result is  $\beta_1 \leq 0$ , indicating non-positive effect on firm productivity.

Table 1: Highlighted Estimation Results

| Dependent Variable: $\hat{\omega}_{it}$ (TFP) | Coefficient | p-value |
|-----------------------------------------------|-------------|---------|
| MHHI $\Delta$                                 | -0.036      | 0.415   |
| ln(Capital)                                   | -0.068***   | 0.000   |
| Bankruptcy Dummy                              | -0.056***   | 0.000   |
| Time Fixed Effects (Base 2014)                |             |         |
| Year 2015                                     | 0.028***    | 0.000   |
| Year 2016                                     | 0.050***    | 0.000   |
| Year 2017                                     | 0.068***    | 0.000   |

Where \*\*\*, \*\*, \* indicate significance at the 1%, 5%, and 10% level.

# What went Wrong

- As DGS established, it is necessary that we establish a correlation between our measure of common ownership and our relevant regressor (TFP)
- Thus, our results suggest that technological innovation is not correlated with common ownership
- Next, we consider two possibilities:
  - ① Model Misspecification - Our model specification misses a key consideration, and so the correlation is lost in our analysis
  - ② Lack of Causal Relationship - In line with DGS, common ownership is not causally related to technological innovation

# Model Misspecification

- ① Missing Controls - Key data such as:
  - D/E Ratios - indication of financial health
  - Region Control - Captures more specific time-invariant effects
  - Fleet Age - To absorb effects of improvement in aircraft manufacturing
- ② Lagging Effect - We assume all effect on productivity are instantaneous (within quarter), however it may be the case that some effects on technology lag
- ③ Nonlinearity of Ownership - It may be that small amounts of common ownership have minuscule effects and there is a critical mass at which common ownership becomes (more) effective

## Lack of Causal Relationship

- ① Investor Passivity - Many institutions take a passive investment strategy, so while there may be common ownership, there is no common voice. Ex: Vanguard's Funds
- ② Price vs Cost Effects - Common owners may facilitate collusion, but firms may still compete in cost efficiency, so even if common ownership is causal for price collusion, it may not be related to productivity collusion
- ③ Offsetting Effects - Assuming common ownership does reduce cost competition, it may also establish opposing productivity forces, Ex: Signaling firm stability, lowering the cost of capital

## What Can Be Concluded

- Our model failed to establish a relationship between common ownership and productivity, however there are many further avenues to explore
- A particularly promising avenue is a more theory-based approach that will inform a better model and allow for a more well-directed exploration

# Thank You!

Questions?